DTMC

Chance

1. PCTL formulas are ϕ with the following syntax:

$$
\begin{array}{lcll}
\phi & ::= & \mathbf{true} \mid p \mid \phi_1 \wedge \phi_2 \mid \neg\phi \mid \mathbf{P}_{\sim p}[\psi] & \text{(state formulas)} \\
\psi & ::= & \mathbf{X}\phi \mid \phi_1 \ \mathbf{U}^{\leq k} \ \phi_2 \mid \phi_1 \ \mathbf{U} \ \phi_2 & \text{(path formulas)}
\end{array}
$$

where p is an atomic proposition, and $\sim$ is one of $\leq, <, \geq, >, =$.

(a) Give the formal semantic of PCTL, interpreted over DTMC.

Answer: Below (see also the slides on DTMC).

Let $M = (S, s_0, P, L)$ be your DTMC. A PCTL formula ϕ holds on M if ϕ holds on the initial state state s_0, denoted by $s_0 \models \phi$. We now give meaning of $s_0 \models \phi$, for any state s. See below:

$$
\begin{array}{lcl}
s \models p & = & p \in L(s) \\
s \models \neg\phi & = & \text{not}(s \models \phi) \\
s \models \phi_1 \wedge \phi_2 & = & s \models \phi_1 \ \text{ and } \ s \models \phi_2
\end{array}
$$

For the probabilistic operator $\mathbf{P}_{\sim p}[\psi]$ the definition is as below. I only give the def. for '$>$' as the $\sim$. The definition is similar for other cases of $\sim$.

$$
s \models \mathbf{P}_{>q}\psi \quad = \quad \underbrace{P\{\ \omega \in Paths(s) \mid \omega \models \psi\ \}}_{\text{probability}} \ > \ q
$$

Here, $Paths(s)$ is the set of all possible paths/executions that starts from s; typically there are infinitely many of them. The notation $\omega \models \psi$ means that the (temporal) property ψ holds on the path ω.

Finally, the notation $P\{\ \omega \in Paths(s) \mid \omega \models \psi\ \}$ refers to the probability that such a path ω occur as the behavior of the model if we start in the state s.

The meaning of $\omega \models \psi$ is defined recursively over the syntax of ψ as follows (the def. is not complete, you can see the slides for the rest):

$$
\begin{array}{lcl}
\omega \models \mathbf{X}\phi & = & \omega[1..] \models \phi \\
\omega \models \phi_1 \ \mathbf{U} \ \phi_2 & = & \omega \models \phi_2 \ \text{ or } \ (\omega \models \phi_1 \text{ and } \omega[1..] \models \phi_1 \ \mathbf{U} \ \phi_2)
\end{array}
$$

The notation $\omega[1..]$ refers to the (infinite) suffix of ω starting from index 1 (so, the sequence $\omega_1, \omega_2, \ldots$.

(b) Let's introduce some derived operators with the 'usual' meaning: $\vee$, $\rightarrow$, and $\mathbf{F}$ (or $\Diamond$). Propose a definition for them.

Answer: This can be defined as 'usual':

$$
\begin{array}{lcl}
\phi_1 \vee \phi_2 & = & \neg(\neg\phi_1 \wedge \neg\phi_2) \\
\phi_1 \rightarrow \phi_2 & = & \neg\phi_1 \vee \phi_2 \\
\mathbf{F}\phi & = & \mathsf{true} \ \mathbf{U} \ \phi
\end{array}
$$

(c) Developer Peter Parker claims that $\neg\mathbf{P}_{>c}\phi$ is equivalent to saying $\mathbf{P}_{\leq c}\phi$. Is Peter right?

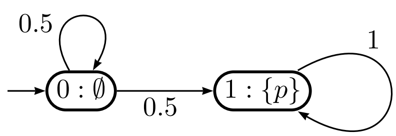
Answer: see in the slides :)

(d) The 'usual' way to define $\mathbf{G}$ is: $\mathbf{G}\phi = \neg\mathbf{F}\neg\phi$. So, for example $\mathbf{P}_{\geq 0.8}[\mathbf{G}a]$ can be expressed as $\mathbf{P}_{\geq 0.8}[\neg\mathbf{F}\neg a]$. Unfortunately, the syntax of PCTL does not have negation over the temporal formula (we only have negation over proposition and probabilistic formulas). So, the formula "$\mathbf{P}_{\geq 0.8}[\neg\mathbf{F}\neg a]$" is not allowed in PCTL. How do you then propose to define $\mathbf{P}_{\geq p}[\mathbf{G}\phi]$?

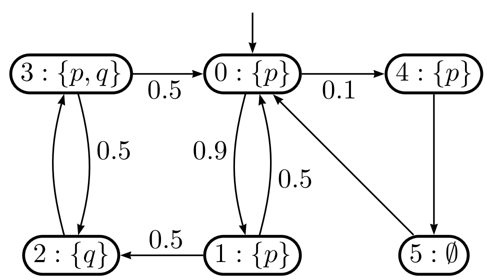
Answer: How about like this: $\mathbf{P}_{\leq p}[\mathbf{F}\neg\phi]$?

2. Explain why in PCTL (e.g. interpreted over DTMC) $\mathbf{P}_{\geq 1}[\mathbf{F}\phi]$ is not the same as CTL $\mathbf{AF}\phi$ (the former is weaker than the latter). Give an example that shows the difference. Note: $\mathbf{F}\phi$ is defined as *true* $\mathbf{U}$ ϕ.

Answer: Consider the DTMC M below. The CTL property $\mathbf{AF}p$ is not a valid property of M, because it can cycle in $S0$ forever, hence never reaching the state $S1$ where p holds. However, from PCTL perspective, the probability that M cycles in $S0$ forever would approach zero. And therefore the probability of eventually going over to $S1$ approaches 1.



3. Consider the following DTMC M modelling some system. $Prop = \{p, q\}$.



Consider the following PCTL properties to check:

- $\mathbf{P}_{\geq 0.82}$ $[p\ \mathbf{U}\ q]$
- $\mathbf{P}_{< 0.82}$ $[p\ \mathbf{U}\ q]$
- $\mathbf{P}_{=?}$ $[p\ \mathbf{U}\ q]$

What do these formula say? Then describe the procedure to check them (or at least one of them) on M.

Answer:

The property $\mathbf{P}_{\geq 0.82}$ $[p\ \mathbf{U}\ q]$ is valid on a state s if the execution that starts from s would satisfy the path formula $\psi = p\ \mathbf{U}\ q$ with the probability of at least 0.82. When the reference state s is not mentioned, we mean it to be M's initial state. Since all executions of M starts from its initial state, this is the same as saying that M's executions satisfy the asserted (probabilistic) property.

The formula $\mathbf{P}_{=?}$ $[p\ \mathbf{U}\ q]$ is a quantitative query (so, not a Y/N-query), asking what the probability of "a state s" to satisfy the path property $p\ \mathbf{U}\ q$. If no reference state is mentioned, then we mean to ask that on s_{init} (well, M's initial state).

Also notice that answering the query $\mathbf{P}_{=?}$ $[p\ \mathbf{U}\ q]$ for every state s in M will enable us to answer the other two queries/formulas above.

Let's now focus on the last formula; let's call the inner formula ψ:

$$\mathbf{P}_{=?}\ [\ \underbrace{p\ \mathbf{U}\ q}_{\psi}\ ]$$

Let us now calculate, for every state, the probability that it satisfies ψ. We will proceed as follows:

(a) We need to first perform "precomputation", to identify the so-called $\mathbf{S}^{yes}$ and $\mathbf{S}^{no}$ sets. We start with $\mathbf{S}^{no}$. This is the set of states of M on which $\mathbf{P}_{\leq 0}[\psi]$ holds. From the picture you can 'see' that this set is $\{4,5\}$, but let us see how we can calculate this systematically.

Calculate first the set E, which is all the states satisfying the (non-probabilistic) CTL formula $\mathbf{E}[p\ \mathbf{U}\ q]$. You can do this e.g. with CTL standard model-checking. We end up with $E = \{0,1,2,3\}$. Any state in E has a path satisfying ψ, and hence a non-zero probability of satisfying ψ. So, $\mathbf{S}^{no}$ must be $S/E = \{4,5\}$.

The set $\mathbf{S}^{yes}$ consists of all states on which $\mathbf{P}_{\geq 1}[\psi]$ holds. To calculate this set, we first calculate the set F of all states t that have a path τ, starting from t, leading to $\mathbf{S}^{no}$, while maintaining the property $p \wedge \neg q$ along τ (up to entering until $\mathbf{S}^{no}$). We end up with $F = \{0,1,4,5\}$[1]. By its definition/construction, states in F has a non-zero probability to satisfy $\neg\psi$. So, $S/F = \{2,3\}$ is the set $\mathbf{P}_{\geq 1}[\psi]$ we look for[2]

So, to summarize we have calculated:

- $\mathbf{S}^{no} = \{4,5\}$.
- $\mathbf{S}^{yes} = \{2,3\}$.

(b) We can now calculate the probability of each state in M for satisfying ψ.

We can conclude that the probability of any state in $\mathbf{S}^{no}$ for satisfying ψ is simply 0. Conversely, the probability of any state in $\mathbf{S}^{yes}$ for satisfying ψ is 1.

For any state t that do not belong to either $\mathbf{S}^{yes}$ or $\mathbf{S}^{no}$, its probability for satisfying ψ is some $0 < p < 1$. We have two such states: $\{0,1\}$. Let x_s denote the probability that a state s satisfies ψ (which is $p\ \mathbf{U}\ q$). We will have the following relations/equations; and x_i's are the solutions of the equations:

$$\begin{aligned} x_0 &= P_{0\to 1} * x_1 + P_{0\to 4} * x_4 \\ x_1 &= P_{1\to 0} * x_0 + P_{0\to 2} * x_2 \end{aligned}$$

where $P_{u\to v}$ denotes the probability of taking the transition to v, if we are in the state u.

Now, we know that $x_4 = 0$ and $x_2 = 1$. So the above simplify to:

$$\begin{aligned} x_0 &= P_{0\to 1} * x_1 \\ x_1 &= P_{1\to 0} * x_0 + P_{0\to 2} \end{aligned}$$

Filling the the values of the P (from M's picture) we obtain these equations:

$$\begin{aligned} x_0 &= 0.9x_1 \\ x_1 &= 0.5x_0 + 0.5 \end{aligned}$$

You can apply algebra to solve the equations. E.g. the first says x_0 is $0.9x_1$. If we fill this in for the x_0 in the 2nd equation we get:

$$x_1 = 0.5 * (0.9x_1) + 0.5$$

This leads to the solution that $x_1 \approx 0.909$. And $x_0 \approx 0.818$.

[1] Notice that $\{4,5\}$ are states from $\mathbf{S}^{no}$ itself. In general, this F will indeed always include $\mathbf{S}^{no}$.

[2] The MC-slides in the website of Prism, F is characterized differently, namely as the set of states where the probability of ψ is < 1. This should be the same as saying that there is a non-zero probability of $\neg\psi$, which is how I looked at it above.

(c) Now we can answer all formulas/queries that were posed in the original question:

- $\mathbf{P}_{=?}$ $[p\ \mathbf{U}\ q]$. On s_{init} the probability is 0.818.
- $\mathbf{P}_{\geq 0.82}$ $[p\ \mathbf{U}\ q]$. Well, then this is **not** valid on s_{init}.
- $\mathbf{P}_{<0.82}$ $[p\ \mathbf{U}\ q]$. This is valid on s_{init}.

4. Consider again the DTMC M in No. 3. On which states of M the following properties hold?

- $\mathbf{P}_{\leq 0}$ $[p\ \mathbf{U}\ \neg(p \vee q)]$.
- $\mathbf{P}_{\geq 1}$ $[p\ \mathbf{U}\ \neg(p \vee q)]$.
- $\mathbf{P}_{\geq 0.18}$ $[p\ \mathbf{U}\ \neg(p \vee q)]$.
- $\mathbf{P}_{<0.18}$ $[p\ \mathbf{U}\ \neg(p \vee q)]$.

5. Consider again the DTMC M in No. 3. Which states satisfy $\mathbf{P}_{\leq 0.5}$ $[p\ \mathbf{U}\ \mathbf{P}_{\geq 1}[q\ \mathbf{U}\ p]]$? (Well, obviously state 5, but which other states, if any, satisfy the property?)

Answer: Let's first introduce some names to abbreviate:

$$\mathbf{P}_{\leq 0.5}\ [\ p\ \ \mathbf{U}\ \ \overbrace{\mathbf{P}_{\geq 1}[\ \underbrace{q\ \mathbf{U}\ p}_{\psi}\ ]}^{\chi}\ ]$$

We will first calculate which states satisfy $\chi = \mathbf{P}_{\geq 1}[\psi]$. We proceed with the checking procedure as in the examples in (3):

(a) We calculate the set E of states satisfying $\mathbf{E}[q\ \mathbf{U}\ p]$, and obtain $E = \{0, 1, 2, 3, 4\}$. So, $\mathbf{S}^{no}$ for ψ is $\{5\}$ (the complement of E).

(b) Calculate the set F consisting states with an outgoing path whose states satisfy $q \wedge \neg p$ and leads to $\mathbf{S}^{no}$. Only the states in $\mathbf{S}^{no}$ itself satisfy this property. So, $F = \{5\}$. $\mathbf{S}^{yes}$ (for ψ) is the complement of F. So, it is $\{0, 1, 2, 3, 4\}$.

(c) From the above we conclude that the probability of satisfying ψ is 1 on all states, except on state 5, for which the probability is 0.

(d) So, the states in $\{0, 1, 2, 3, 4\}$ satisfy $\mathbf{P}_{\geq 1}[\psi] = \chi$.

Next, we proceed to calculate the probabilities of different states to satisfy $p\ \mathbf{U}\ \chi$. From this we can see which states would satisfy $\mathbf{P}_{\leq 0.5}$ $[p\ \mathbf{U}\ \chi]$ (which abbreviates $\mathbf{P}_{\leq 0.5}$ $[p\ \mathbf{U}\ \mathbf{P}_{\geq 1}[q\ \mathbf{U}\ p]]$).

The calculation proceeds in the same way:

(a) We calculate the set E of states satisfying $\mathbf{E}[q\ \mathbf{U}\ \chi]$. Treat χ as an atomic proposition. Previously we already calculated which states satisfy χ, namely $\{0, 1, 2, 3, 4\}$; so, add χ as a 'label' to these states.

We then obtain that E is also $\{0, 1, 2, 3, 4\}$. So, $\mathbf{S}^{no}$ (for $q\ \mathbf{U}\ \chi$) is $\{5\}$.

(b) Calculate the set F consisting states with an outgoing path whose states satisfy $q \wedge \neg\chi$ and leads to $\mathbf{S}^{no}$. Only the states in $\mathbf{S}^{no}$ itself satisfy this property. So, $F = \{5\}$. $\mathbf{S}^{yes}$ is the complement of F. So, it is $\{0, 1, 2, 3, 4\}$.

(c) From the above we conclude that the probability of satisfying $q\ \mathbf{U}\ \chi$ is 1 on all states, except on state 5, for which the probability is 0.

(d) So, the only state satisfying the asked property $\mathbf{P}_{\leq 0.5}$ $[p\ \mathbf{U}\ \chi]$ is state 5.

6. PLTL formulas ϕ has the following syntax: where p is an atomic proposition, and $\sim$ is one of $\leq, <, \geq, >, =$.

$$
\begin{array}{lcll}
\phi & ::= & \mathbf{P}_{\sim p}[\psi] & \\
\psi & ::= & p \mid \psi_1 \wedge \psi_2 \mid \neg\psi \mid \mathbf{X}\psi \mid \psi_1 \;\mathbf{U}\; \psi_2 & \qquad \text{(path formulas)}
\end{array}
$$

Note that a path formula can only recurse over other path formulas. In particular, unlike in PCTL, a PLTL path formula cannot have a probabilistic sub-formula.

Give the formal semantic of PLTL, interpreted over DTMC.

Answer: Let $M = (S, s_0, P, L)$ be your DTMC. A PLTL formula ϕ holds on M if ϕ holds on the initial state state s_0, denoted by $s_0 \models \phi$. In PLTL, ϕ takes the form of $\mathbf{P}_{\sim p}[\psi]$, for some path formula ψ. For any state s, the definition of $s \models \mathbf{P}_{\sim p}[\psi]$ is given below, for '>' as the $\sim$. The definition is similar for other cases of $\sim$.

$$
s \models \mathbf{P}_{>q}\psi \quad = \quad \underbrace{P\{\, \omega \in \mathit{Paths}(s) \mid \omega \models \psi \,\}}_{\text{probability}} \; > \; q
$$

Here, $\mathit{Paths}(s)$ is the set of all possible paths/executions that starts from s; typically there are infinitely many of them. The notation $\omega \models \psi$ means that the (temporal) property ψ holds on the path ω.

Finally, the notation $P\{\, \omega \in \mathit{Paths}(s) \mid \omega \models \psi \,\}$ refers to the probability that such a path ω occur as the behavior of the model if we start in the state s. This part is the same as what we had in the solution of question No. 1 on PCTL.

Since ψ is a purely LTL formula, the meaning of $\omega \models \psi$ is as standard in LTL. For your convenience this is shown below, using the structure as in Kwiatkowska-Parker (rather than the structure I used in the Lecture Notes):

$$
\begin{array}{lcl}
\omega \models p & = & p \in L(\omega_0) \\
\omega \models \neg\psi & = & \text{not} \;\; \omega \models \psi \\
\omega \models \psi_1 \wedge \psi_2 & = & \omega \models \psi_1 \;\; \text{and} \;\; \omega \models \psi_2 \\
\omega \models \mathbf{X}\psi & = & \omega[1..] \models \psi \\
\omega \models \psi_1 \;\mathbf{U}\; \psi_2 & = & \omega \models \psi_2 \;\; \text{or} \;\; (\omega \models \psi_1 \text{ and } \omega[1..] \models \psi_1 \;\mathbf{U}\; \psi_2)
\end{array}
$$

7. Given an example that demonstrates that PLTL formula $\mathbf{P}_{\geq 1}[F\psi]$ is not the same as plain LTL formula $\Diamond\psi$.

Answer: The same example as in No. 2. The PLTL $\mathbf{P}_{\geq 1}[Fp]$ holds on the DTMC there, but the plain LTL $\Diamond p$ is not valid on the model.

8. Consider again the DTMC M in No. 3.

   (a) What are the SCCs of M? Which ones are bottom SCCs (BSCCs)?

   Answer: There is only one SCC: the entire M. Therefore it is also a BSCC.

   (b) Does the PLTL property $\mathbf{P}_{\geq 1.0}[\mathbf{GF}(p \wedge q)]$ hold on M?

   Answer: Property of the form $\mathbf{GF}a$ where a is atomic can be checked directly via BSCCs. As mentioned above, the entire M is the one BSCC we have. Some states in M satisfies $p \wedge q$. So, $\mathit{Prob}(s, p, \mathbf{GF}(p \wedge q))$ from any state s in M is 1, including the initial state 0. In other words: yes, $\mathbf{P}_{\geq 1.0}[\mathbf{GF}(p \wedge q)]$ holds on M.

You might argue that within the BSCC M, the program could cycle e.g. $0-1-0$... forever, and hence avoiding $p \wedge q$. But note that the probability that this cycle occurs approaches 0. And similarly so for other cycles within the BSCC that avoids state 3 where $p \wedge q$ holds.

(c) How about $\mathbf{P}_{>0}[\mathbf{FG}p]$?

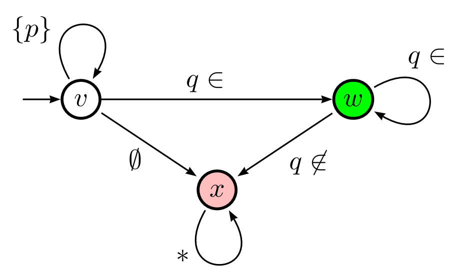
Answer: Property of the form $\mathbf{FG}a$ where a is atomic can also be checked directly via BSCCs. As said above, we only have one BSCC, which is the whole M. Not all states in this BSCC have p hold on them, e.g. p does not hold on state 5. So, $Prob(s, p, \mathbf{FG}p)$ from any state s in M, including the initial state 0, is zero. In other words: $\mathbf{P}_{>0}[\mathbf{FG}p]$ does *not* hold on M.

You might argue that within the BSCC M, the program could cycle $0-1-0$... forever, which has the property $\mathbf{FG}p$. But as pointed out before, the probability that this cycle occurs approaches 0.

9. Consider again the DTMC M in No. 3. We want to check if it satisfies the PLTL property $\mathbf{P}_{\leq 0.8}[(p \wedge \neg q)\ \mathbf{U}\ \mathbf{G}q]$.

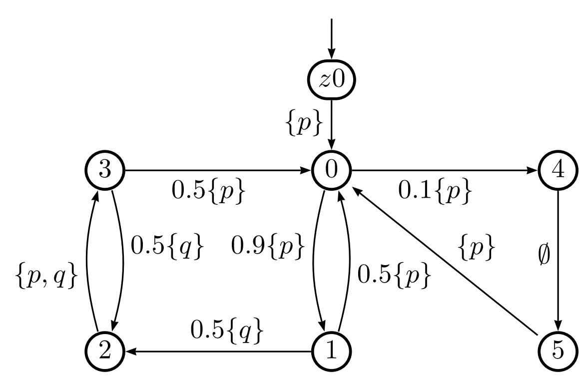
Perform the PLTL model checking algorithm to check this.

Answer: The LTL property $\psi = (p \wedge \neg q)\ \mathbf{U}\ \mathbf{G}q$ can be expressed by the following deterministic Rabin automaton R:

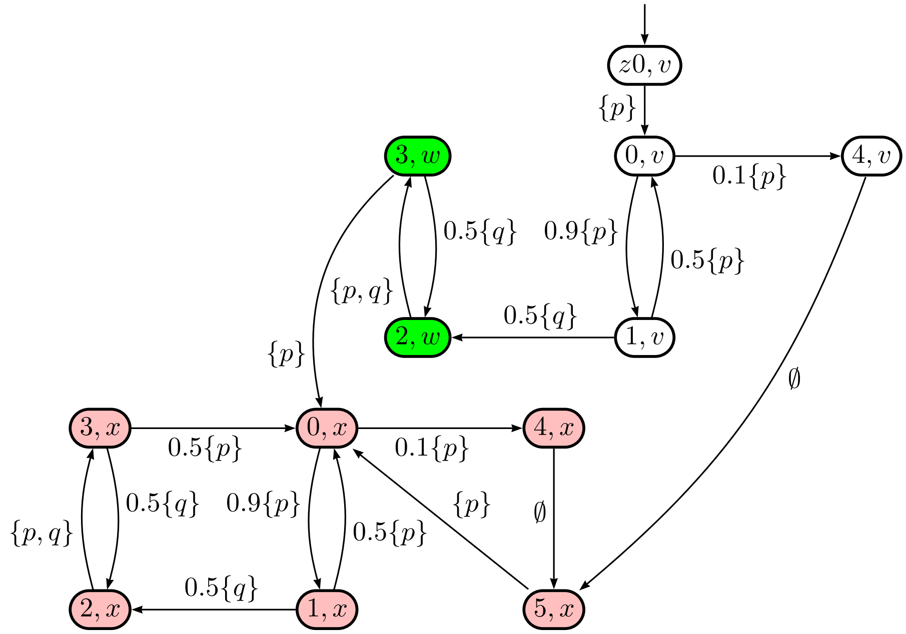


The accepting condition is $\{(L, K)\}$ where $L = \emptyset$ and $K = \{w\}$. Note: well, since L is empty, it is also a Buchi. The red-states are some kind of sink states, once there you can't get ψ_a anymore.

Before we construct the intersection $M \cup R$, let's first transform M by moving the labels to incoming arrows. We add a fake initial state too:



The intersection $M \cap R$ is shown below:



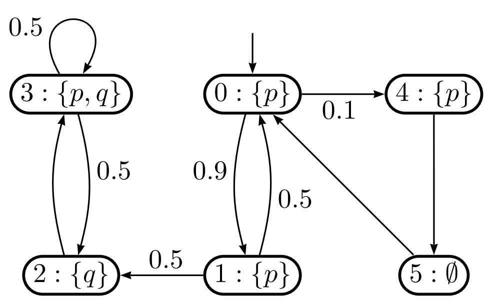
The accepting condition of $M \cap R$ is $\{(L, K)\}$ where L is empty (because the L of R was empty) and $K = \{(2, w), (3, w)\}$ (the green states).

Pfff... that was a big one. But anyway, there is only one bottom SCC in $M \cap R$, formed by the red-states. Let's call this BSCC Z. This Z does not contain a member of K, so it is not accepting. As this is the only BSCC that we have, we have no accepting execution of $M \cap R$ either.

So, $Prob(z0, \psi)$ is zero. Note that this does not necessarily mean that an execution in M with property ψ is impossible (in fact, it is possible); it's just that the probability of having that approaches 0 on limit.

So, the PLTL property $\mathbf{P}_{\leq 0.8}[(p \wedge \neg q)\ \mathbf{U}\ \mathbf{G}q]$ does not hold on M.

10. Consider the following variation of the DTMC M in No. 3. The *Prop* is the same, namely $\{p, q\}$. The original transition $3 \to 0$ has been replaced with a self-transition $3 \to 3$.

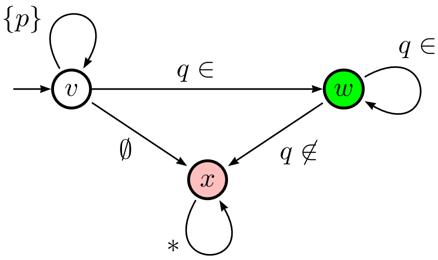


We want to check if this new M satisfies the PLTL properties:

- $\phi_a \ = \ \mathbf{P}_{\leq 0.8}[(p \wedge \neg q)\ \mathbf{U}\ \mathbf{G}q]$.
- $\phi_b \ = \ \mathbf{P}_{\leq 0.8}[p\ \mathbf{U}\ \mathbf{G}q]$.

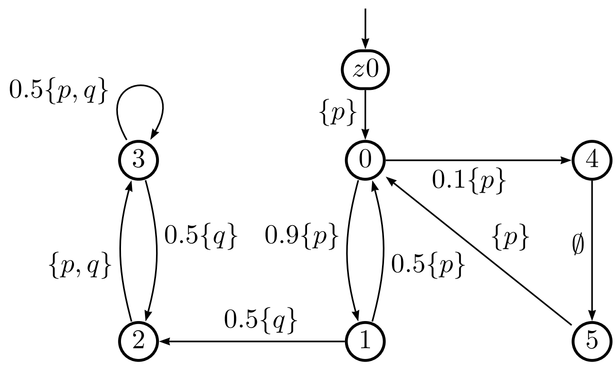
Perform the PLTL model checking algorithm to check them.

Answer: Let's first do (a). The LTL property $\psi_a = (p \wedge \neg q)\ \mathbf{U}\ \mathbf{G}q$ is the same as what we had in question No. 9. Recall that it can be expressed by the following deterministic Rabin automaton; here let's call it Ra:

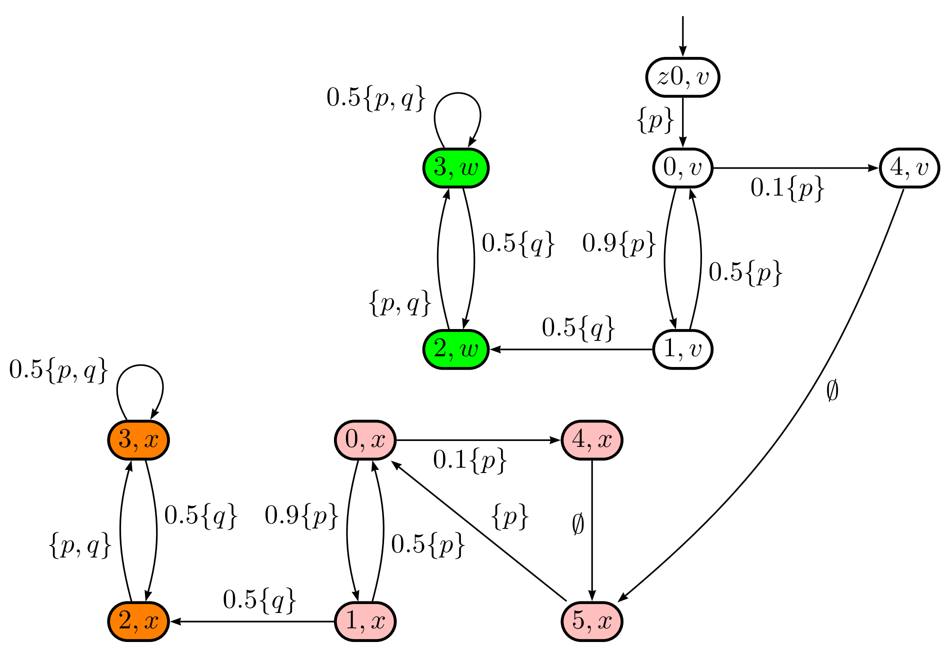


The accepting condition is $\{(L, K)\}$ where $L = \emptyset$ and $K = \{w\}$. The red-states are sink states, once there you can't get ψ_a anymore.

Before we construct the intersection $M \cap Ra$, let's first transform M by moving the labels to incoming arrows. We add a fake initial state too:



The intersection $M \cup Ra$ is shown below:



The accepting condition of $M \cap Ra$ is $\{(L, R)\}$ where L is empty (because the L of Ra was empty) and $K = \{(2, w), (3, w)\}$ (the green states).

There are two bottom SCCS: Z_1 consisting of the green states, and Z_2 consisting of the orange states. Z_1 is accepting as $Z_1 \cap K$ is non-empty and $Z_1 \cup L$ is empty.

Z_2 is non-accepting because $Z_2 \cap K$ is empty.

Any execution (from the initial state) satisfying ϕ_a will end up in an accepting BSCC; in our case, ending up in Z_1. So the probability $Prob((z0, v), \phi_a)$ can be calculated via $Prob((z0, v), \mathbf{F}Z_1)$ (eventually reaching the BSCC Z_1).

Obviously, the probability of having $\mathbf{F}Z_1$ from any state in Z_1 itself is 1.

The probability of having $\mathbf{F}Z_1$ from any state containing x (red and orange states) is zero. x was a sink state in Ra, from where you have no route to K.

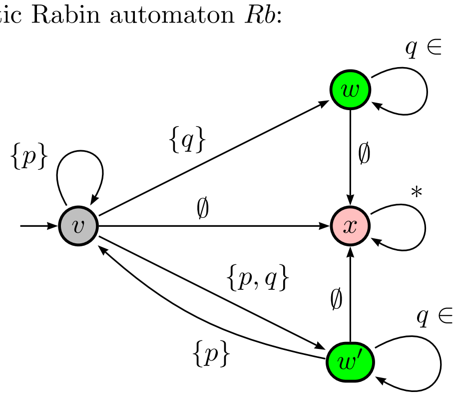
- From any state τ in $Z1$, $Prob(\tau, \mathbf{F}Z_1) = 1$.
- From any red or orange state τ $Prob(\tau, \mathbf{F}Z_1) = 0$.
- $Prob((z0, v), \mathbf{F}Z_1) = Prob((0, v), \mathbf{F}Z_1)$.
- $Prob((0, v), \mathbf{F}Z_1) = 0.1 * Prob((4, v), \mathbf{F}Z_1) + 0.9 * Prob((1, v), \mathbf{F}Z_1)$.
- $Prob((4, v), \mathbf{F}Z_1) = Prob((5, x), \mathbf{F}Z_1) = 0$
- $Prob((1, v), \mathbf{F}Z_1) = 0.5 * Prob((0, v), \mathbf{F}Z_1) + 0.5 * Prob((2, w), \mathbf{F}Z_1)$.

Simplifying the equations give: $Prob((1, v), \mathbf{F}Z_1) = 0.5*0.9*Prob((1, v), \mathbf{F}Z_1) + 0.5*1$. So, $Prob((1, v), \mathbf{F}Z_1) = 50/55$.

So, $Prob((z0, v), \mathbf{F}Z_1) = Prob((0, v), \mathbf{F}Z_1) = 0.818....$

So, the PLTL property $\mathbf{P}_{\leq 0.8}[(p \wedge \neg q)\ \mathbf{U}\ \mathbf{G}q]$ does not hold on M.

For question (b): the LTL property $\psi_b = p\ \mathbf{U}\ \mathbf{G}q$ can be expressed by the following deterministic Rabin automaton Rb:



Accepting condition: $\{(L, K)\}$ where $L = \{v\}$ and $K = \{w, w'\}$. The red state is a sink state.

Although Rb is certainly different from the previous Ra, the intersection $M \cap Rb$ gives (accidentally) the same resulting automaton as the previous intersection $M \cap Ra$, with the same states, the same initial state, the same transitions, and the same probability-numbers. The accepting condition is slightly different though: $\{(L, K)\}$ with K containing the green states and L containing all the states with v (non-colored states).

Try to construct it yourself, and see what you get.

So... what do you think? Does $\phi_b = \mathbf{P}_{\leq 0.8}[p\ \mathbf{U}\ \mathbf{G}q]$ hold on M?